OIT367 - Business Intelligence from Big Data

Spring 2023

Final Project Report

Predicting Employees' "Willingness To Stay" at Springboard

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You can keep our project for future use

Executive Summary

Problem

Springboard is an e-learning platform that prepares people for modern, tech careers through cutting edge curriculum and mentor guidance. Springboard currently runs a pulse survey twice a year that asks employees about the company culture and their career outlook. We were tasked by Springboard to run a regression analysis to predict employees' willingness to stay using responses from the survey data.

Principal Findings

The regression analysis of employee survey data provided several insights for Springboard to explore further. Given we were not able to tie actual turnover information to the survey responses, we have used the response to "*I see myself working at Springboard a year from now*" as a proxy for willingness to stay. Two survey questions and respective employee responses were identified as most predictable indicators of employees' willingness to stay: *I am learning & growing my skill set in my current role* and *On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*

More specifically, employees responding "neutral" to their self-perceived learning curve or growth was negatively correlated with their willingness to stay while likelihood of referring Springboard to a friend was a positive indicator of their intention to stay. Despite the statistical significance of these two indicators, it is important to note that the causality of these findings is not conclusive.

Recommendations

Springboard could conduct additional research to understand what motivates employees to make referrals and experiment with incentives or mechanisms to make referrals easier or more attractive for employees. Springboard could also improve its learning and development offerings and encourage employees to be more invested in these opportunities.

In addition, the small sample size of the survey and the reliance on self-reported attitudes limit the robustness of the analysis. Incorporating more objective measures such as employee performance and actual referral data could provide more actionable insights.

Despite these limitations, the analysis provides a starting point for Springboard to take action and further explore the relationship between workplace data and retention.

Introduction

We were tasked by Springboard to run a regression analysis using responses from the survey and workplace data to identify which survey questions are most predictive of employees' willingness to stay. Our insights and findings will be directly relevant to the HR and executive team but, more broadly, can also be informative to other companies of similar size, industry, or geography in determining what questions and metrics to track in the regular employee surveys that relate to retention and attrition. See ***Appendix A*** for additional background.

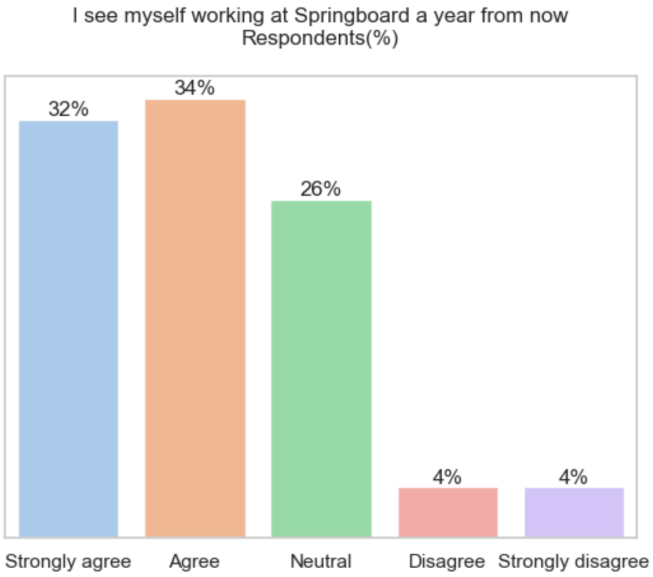
Data

Our data was sourced directly from Springboard's most recent pulse survey from May 2022. It contains 180 responses, representing a 91% response rate for Springboard's employees and contractors. They represent a variety of different teams, mainly based in India and San Francisco, with the largest ones being Sales, Technology, and People Ops (***Appendix B***).

The dataset contains 48 columns, including a timestamp for completion and responses for 47 survey questions. Survey questions revolved around company culture, work-life balance, other topics related to the general work environment, as well as expectations regarding working at Springboard in a year (see ***Appendix C*** for full list). Response types range from categorical variables to free-form text answers. The former includes items such as agreement / disagreement scales (Strongly Agree to Strongly Disagree) and views on company values. The only numerical variable was one that asked how likely respondents were to refer Springboard to a friend on a scale of 1 to 10.

Given we were not able to tie actual turnover information to the survey responses, we have used the response to "*I see myself working at Springboard a year from now*" as a proxy for willingness to stay and hence the variable we are looking to predict. Most respondents were positive with 66% saying they strongly agree or agree with the statement; 8% were negative but a larger subset (26%) were neutral on their outlook.

Exhibit 1: Distribution of Responses for "*I see myself working at Springboard a year from now*"



Methodology

0. Model Selection

Given our outcome variable (i.e., response to "*I see myself working at Springboard a year from now*") is a 5-point categorical scale, we had to decide between a binary classification model or a continuous regression model. We ultimately built a linear regression model given the data distribution as well as the applicability and simplicity of the model.

With a binary classification model (e.g, logistic regression), we would have had to bucket the responses into two categories: "willing to stay" and "not willing to stay." This created the issue of determining whether the "neutral" response would go into the "willing to stay" or the "not willing to stay" category. Including it in the "willing to stay" category would have left only 8% of respondents in the other category within an already small sample size. Conversely, including it in the "not willing to stay" category would increase the chance of false negatives where people who were not intending to leave Springboard would be flagged as flight risks. The output would create two buckets of low and high flight risk employees without providing more nuanced insight into the level of risk.

A linear regression would simplify this decision and empower Springboard to determine their risk tolerance. The linear regression output can provide Springboard with a ranked order list of high to low willingness to stay employees. Based on their internal capacity and priorities, Springboard could determine at which point they want to continue or stop intervening with high-risk employees.

1. Data Preparation

We first cleaned up and processed the data to prepare it for model building. The data was randomized in advance of creating test and training data sets. We removed 5 columns, including a timestamp and 4 open-ended questions (e.g., what do you think is Springboard's cilantro? something Springboarders love but others might dislike). The timestamp was not relevant and the open-ended responses would require more nuanced sentiment analysis outside of the scope for this project. Additionally, many responses to the open-ended questions were "None/NA" or ones that expressed confusion about the questions.

Next, we replaced missing values with the mode of each question. Overall, the data was fairly clean and consistent given the limited set of survey responses each respondent could select from.

We then binned the categorical variables[1] and reassembled predictors to form a comprehensive dataset. Finally, we renamed the column names (from "F1" to "F173").

Last, we converted the values for the outcome, "*I see myself working at Springboard a year from now,*" to a numerical scale:

1	2	3	4	5
Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

[1] Creating m-1 binary dummy variables for a categorical with m different values

2. Feature Set Exploration & Selection

A) LASSO Cross-Validation (CV) with standalone features

First, we utilized the LASSO regularization method with 5[2]-fold CV on all features in the dataset. There were six features with non-zero coefficients (***Appendix D***):

F8	*Mark the one Springboard value you think... [We, as a company, exhibit most consistently]* = Mission First
F52	*I am empowered to make decisions within my role* = Strongly agree
F55	*I am learning & growing my skillset in my current role* = Neutral
F64	*Hiring a diverse group of candidates is a priority to leaders at Springboard* = Neutral
F150	*My preference is to work [x] out of the SF or BLR office* = 1 week/quarter
F173	*On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*

B) LASSO CV with standalone features and two-way interaction terms

To explore potential relationships between the features, we also generated two-way interaction terms and used the LASSO regularization method on the individual features and interactions. There were two features with non-zero coefficients, both of which were interaction terms, (***Appendix E***):

F55_F173	Interaction between *I am learning & growing my skillset in my current role = Neutral* and *On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*
F173_F173	Squared value of the response to the question *"On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?"*

3. Model Building and Comparison

We compared six different regression models using 5-fold CV. RMSE was our primary performance indicator.

A. outcome ~ *(all individual features) + (all interactions with F173)*
   - Given the large number of potential features from interaction terms that requires significantly more computational power, we choose to explore only interactions with F173[3] since the LASSO analysis showed that it would be most relevant
B. outcome ~ *all individual features only*
C. outcome ~ F8[4] + F52[5] + F55[6] + F64[7] + F150[8] + F173[3]
   - Features with non-zero coefficients from the LASSO with only standalone features
D. outcome ~ F55[6]_F173[3] + F173[3]_F173[3]
   - Features with non-zero coefficients from the LASSO with standalone and interactions
E. outcome ~ F8[4] + F52[5] + F55[6] + F64[7] + F150[8] + F173[3] + F55[6]_F173[3] + F173[3]_F173[3]
   - Features with non-zero coefficients from both LASSO models
F. outcome ~ F55[6] + F173[3] + F55[6]_F173[3] + F173[3]_F173[3]
   - Interaction terms from model D and their standalone features

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[2] We have used 5 folds given the relatively small sample size; one can consider increasing the number of folds for larger sample sizes
[3] F173: *On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*
[4] F8: *Mark the one Springboard value you think... [We, as a company, exhibit most consistently]*" = Mission First
[5] F52: *I am empowered to make decisions within my role* = Strongly agree
[6] F55: *I am learning & growing my skillset in my current role* = Neutral
[7] F64: Hiring a diverse group of candidates is a priority to leaders at Springboard = Neutral
[8] F150: *My preference is to work [x] out of the SF or BLR office* = 1 week/quarter

Model C produced the best result (lowest average validation RMSE and standard error).

Exhibit 2: RMSE Comparison from 5-Fold CV

	modelA	modelB	modelC	modelD	modelE	modelF
0	2.597943	2.156764	0.654541	0.670635	0.651392	0.661639
1	2.228047	2.640294	0.708572	0.712215	0.723124	0.730702
2	2.190593	2.411273	0.953729	0.964039	0.952670	0.970837
3	2.439909	2.237514	0.923805	0.983678	1.086768	1.087356
4	2.142364	1.945938	0.649077	0.650825	0.662834	0.672416

```
Model A avg RMSE = 2.3198  and roughly estimated SE = 0.077
Model B avg RMSE = 2.2784  and roughly estimated SE = 0.105
Model C avg RMSE = 0.7779  and roughly estimated SE = 0.0596
Model D avg RMSE = 0.7963  and roughly estimated SE = 0.0655
Model E avg RMSE = 0.8154  and roughly estimated SE = 0.0778
Model F avg RMSE = 0.8246  and roughly estimated SE = 0.0772
```

Results (Model C)

A) Model Summary[9] (see **Appendix E** for more detail)

Least Squares Regression Model: ***outcome***[10] ***~ F8 + F52 + F55 + F64 + F150 + F173***

Exhibit 3: Summary of Linear Regression Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                outcome   R-squared:                       0.436
Model:                            OLS   Adj. R-squared:                  0.411
Method:                 Least Squares   F-statistic:                     17.66
Date:                Wed, 15 Mar 2023   Prob (F-statistic):           4.40e-15
Time:                        14:13:27   Log-Likelihood:                -172.14
No. Observations:                 144   AIC:                             358.3
Df Residuals:                     137   BIC:                             379.1
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.7664      0.353      2.171      0.032       0.068       1.465
F8            -0.2214      0.156     -1.417      0.159      -0.530       0.088
F52           -0.1295      0.142     -0.913      0.363      -0.410       0.151
F55           -0.4969      0.185     -2.680      0.008      -0.864      -0.130
F64            0.1993      0.152      1.307      0.193      -0.102       0.501
F150          -0.2687      0.153     -1.753      0.082      -0.572       0.034
F173           0.3996      0.041      9.705      0.000       0.318       0.481
==============================================================================
Omnibus:                       44.533   Durbin-Watson:                   2.105
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              101.228
Skew:                          -1.300   Prob(JB):                     1.04e-22
Kurtosis:                       6.180   Cond. No.                         44.4
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
validation RMSE= 0.6490768607914658
```

[9] Based on the last training and validation set from the 5-fold cross-validation conducted
[10] "*I see myself working at Springboard a year from now*"; scale of 1 to 5 where Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly agree = 5

B) Interpretation of Statistically Significant[11] Features

1. "*I am learning & growing my skill set in my current role"* = Neutral: Outcome[12] (scale of 1-5) decreases by -0.4969 if the employee's response is neutral
2. "*On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?"*: Outcome[12] (scale of 1-5) increases by 0.3396 for a one unit increase in likelihood score to recommend the company to a friend (scale of 0-10)

Given the small dataset and analysis on a proxy for "willingness to stay", one should focus on the nature of the relationship rather than specific magnitude of the coefficients.

Exhibit 4: Outcome vs feature visualization - F55 and F173

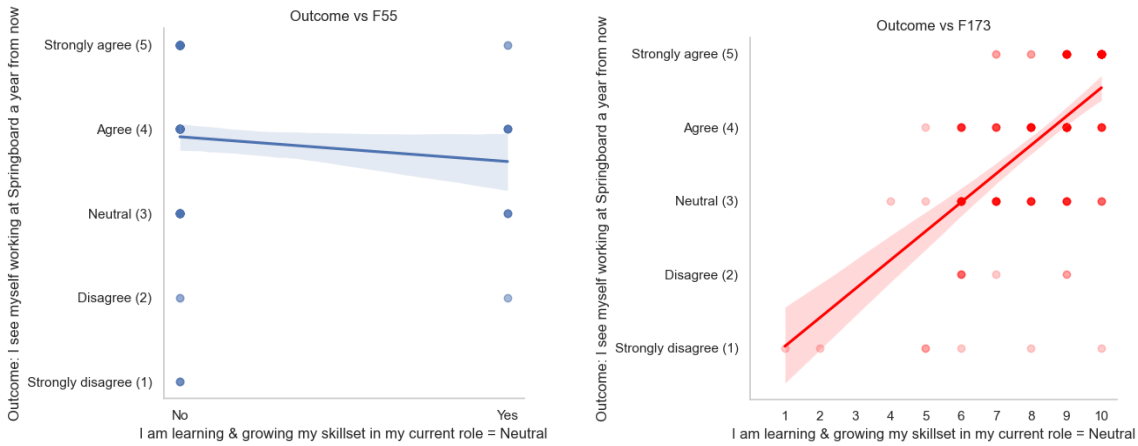
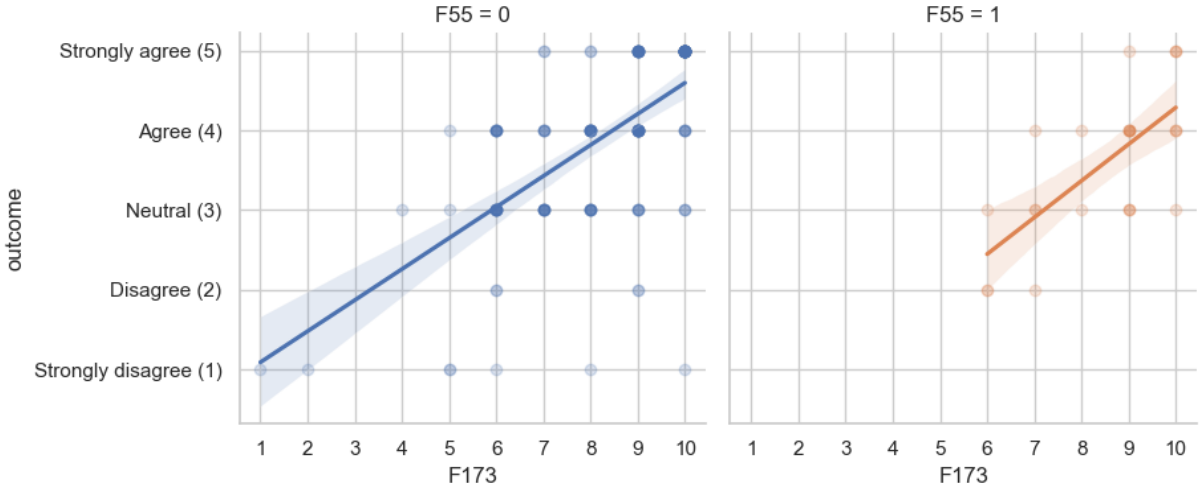


Exhibit 5: Outcome vs F173 visualization by F55 (0 and 1)



[11] At a 5% significance level
[12] "*I see myself working at Springboard a year from now*"; Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly agree = 5

Discussion

The results provide insights that Springboard can begin to take action on or explore further. Likelihood to refer Springboard to a friend was a common predictor from our multiple LASSO models. Intuitively, it makes sense that those who would endorse Springboard to their friends would be more likely to stay at the company. Springboard could conduct additional research and interviews to understand what makes employees more or less willing to recommend Springboard to their friends. Springboard could also consider experimenting with incentives or mechanisms that make referrals easier or more attractive for employees.

An answer of "neutral" towards being able to grow and learn in their current role was a statistically significant predictor of lower willingness to stay. This may be because those who have a positive attitude want to stay and those who stated "disagree" or "strongly disagree" feel more comfortable voicing their opinions to their managers and are interested in working to improve their situation. Perhaps those who feel ambivalent are the most discouraged about their future at Springboard. Springboard could look to improve their learning/development offerings and encourage employees to be more invested in these opportunities.

However, it is important to note that these features are correlated with willingness to stay but their causality is not conclusive. It is not guaranteed that encouraging employees to refer Springboard to their friends will make them more likely to stay. The relationship could be causal either because employees start to buy into Springboard's benefits the more they vocalize them to others or these referrals actually get their friends to join Springboard, making it a more pleasant work environment. Yet instead, there could be a confounding variable, another factor not captured in the model but influencing willingness both to stay and to refer friends. In the latter case, stimulating employee referrals would not improve employee retention. More data and experimentation is required to understand the cause and effect.

The remaining attributes are less actionable and are not statistically significant. For example, knowing that employees who think Springboard best exhibits its value of Mission First are less willing to stay is not very actionable. Advising Springboard to be less mission-driven is not a recommendation that we would make. Additionally, those strongly agreeing that they are empowered to make decisions were found to have a negative relationship with willingness to stay. This contradicts our natural intuition but this could just be noise given the effect is small (coefficient = -0.1295) and not statistically significant (p-value = 0.363).

In addition to the causality issue, there are further caveats to our analysis. First, the small sample size of 180 responses against 40+ predictor variables was a challenge. Leveraging data from prior surveys to increase sample size could be helpful for the next iteration to improve the model's robustness.

Another caveat is that self-reported willingness to stay is not necessarily a proxy indicative of actual employee retention. Using real employment data on employee attrition would make the models more valid in predicting an outcome that has real business implications. Linking survey responses with employment data may be difficult given the survey captures responses anonymously, but there may be methods to combine the data while protecting anonymity.

Finally, the available data largely depends on employees' self-reported attitudes. Incorporating less subjective measures (e.g., salary, employee performance, demographic data) could make the model more robust and actionable. Adding employee performance data could help Springboard

determine at which employees they should target their retention efforts based on both employee flight-risk and their impact to the business.

Conclusion

Following our results, there are several takeaways we would like to share:

1. Two survey questions and respective employee responses were found to be statistically significant and most predictive of employees' willingness to stay:
    - *I am learning & growing my skill set in my current role*
    - *On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*

    Employees responding "neutral" to their self-perceived learning curve or growth was negatively correlated with their willingness to stay while likelihood of referring Springboard to a friend was a positive indicator of their intention to stay. Despite the statistical significance of these two indicators, it is important to note that the causality of these findings is not conclusive.
2. Based on the identified questions and their predictive relationship with employees' reported willingness to stay, Springboard could conduct additional research and interviews to further explore and understand what motivates employees to make referrals as well as offerings related to employee learning and development. We recommend Springboard to experiment with incentives such as referral bonuses or mechanisms that make referrals easier or more attractive for employees.
3. Given the intended objective of the HR and executive team to identify signals for employee retention, it is important to note that the casualty of our findings is not conclusive. For example, encouraging employee referrals may not necessarily improve the actual retention, and other factors not captured in the model may influence employees' reported willingness to stay. In addition, the small sample size of the survey and the reliance on self-reported attitudes limit the robustness of the analysis. Incorporating more objective measures such as employee performance and actual referral data could provide more actionable insights.

Appendix

Appendix A: Additional Background Information

Springboard, founded in 2013 and headquartered in San Francisco, is a company offering an online learning platform that prepares students for the tech industry's most sought and in-demand careers. With expert mentorship from top professionals and a comprehensive curriculum, it provides a network of support and equips students with necessary skills and capabilities to excel in software engineering, data science, UI/UX design, cybersecurity, tech sales, and more.

Springboard's HR and executive team were initially interested in analyzing and predicting employee turnover using the workplace variables and data. It runs a pulse survey twice a year where one of the questions that it asks its employees is about their "intent to stay." Given the available data and questions asked on the survey, they wanted to investigate what the most predictive questions were related to the employees' willingness to stay. Identifying which signals best predict willingness to stay will help them to understand what parts of the company culture and practices they can focus their efforts and resources on.

Appendix B: Respondent Location Base and Team

Which team are you on? / Where is your base?	Course Talent Ops	Finance, IT, Biz Ops, Project Management	Marketing & GMs	People Ops, Office of the CEO, B2B	Product, CourseDev, Design	Program Ops Other (Career Outcomes, LEO, Support, Community and Alumni)	Sales	Student Success	Technology	University Partnerships	All
Elsewhere	2	5	8	6	7	1	18	6	1	11	65
India	0	0	1	5	3	2	0	0	31	4	46
San Francisco	3	10	5	10	8	15	10	7	0	1	69
All	5	15	14	21	18	18	28	13	32	16	180

Appendix C: Complete List of Variables Provided

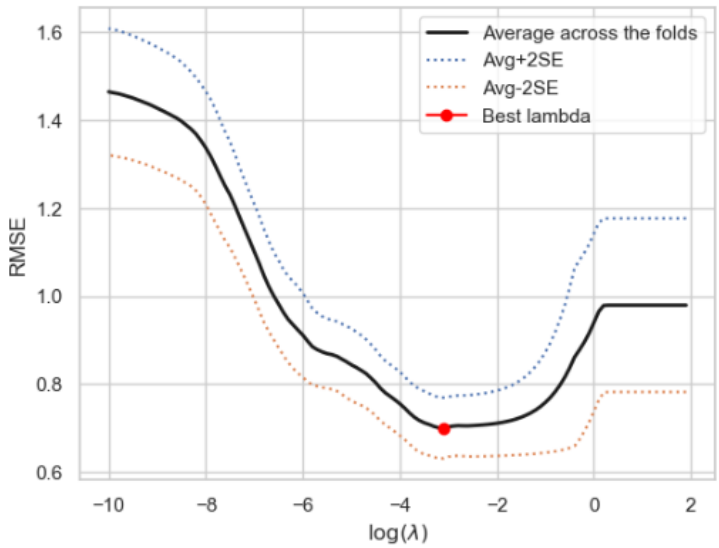
1. I understand how my day to day activities fit into Springboard's strategy and contribute to the mission
2. Mark the one Springboard value you think... [We, as a company, exhibit most consistently]
3. Mark the one Springboard value you think... [We, as a company, exhibit least consistently]
4. Mark the one Springboard value you think... [Our execs exhibit most consistently]
5. Mark the one Springboard value you think... [Our execs exhibit least consistently]
6. Generally, I believe my workload is reasonable for my role
7. I believe I can take time off & disconnect
8. I see myself working at Springboard a year from now
9. I enjoy the majority of what makes up my day to day work
10. My strengths are utilised within my team
11. I know what constitutes good performance in my role
12. I am empowered to make decisions within my role
13. I am learning & growing my skillset in my current role
14. The iGrow program helped me & my manager have meaningful career conversations
15. Hiring a diverse group of candidates is a priority to leaders at Springboard
16. Springboard has a climate in which diverse perspectives are valued
17. I feel comfortable giving feedback to my colleagues
18. I feel comfortable giving feedback to my manager
19. I feel comfortable giving feedback to my exec

20. I feel connected to my team
21. I feel like I belong at Springboard
22. If I made a mistake, I feel like it would be easy to regain my credibility in the eyes of my manager
23. I believe my total compensation (base salary, benefits and equity) is fair
24. The performance review process at Springboard is fair and objective
25. This review cycle minimized manual work for review participants
26. The overall review cycle timelines were...
27. The following programs make me feel valued and appreciated as a Springboard employee [iGrow]
28. The following programs make me feel valued and appreciated as a Springboard employee [OTJT]
29. The following programs make me feel valued and appreciated as a Springboard employee [Learning Budget]
30. The following programs make me feel valued and appreciated as a Springboard employee [Sabbatical]
31. The following programs make me feel valued and appreciated as a Springboard employee [Modern Health]
32. The following programs make me feel valued and appreciated as a Springboard employee [Employee Referral Program]
33. The following programs make me feel valued and appreciated as a Springboard employee [Anniversary Program]
34. The following programs make me feel valued and appreciated as a Springboard employee [Annual Company Offsites]
35. What do you think is Springboard's cilantro?(something Springboarders love but others might dislike) e.g. black liquorice or pineapple on a pizza
36. On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?
37. If your score is anything between 0 - 8, what would it take to move your rating to 9 or 10?
38. My preference is to work [x] out of the SF or BLR office
39. What changes would make you feel better about Springboard's approach to hybrid work?
40. Do you have any additional feedback to share about your organization (e.g. BizOps, Technology, Sales, etc) and/or about Springboard?
41. Where is your base?
42. Are you a contractor or employee?
43. How long have you been at Springboard?
44. Are you an individual contributor or a manager?
45. Which team are you on?
46. Did you attend the Global Offsite in Santa Cruz or India team quarterly meetup?
47. If you'd like to have a longer conversation about your feedback, please share your email.

Appendix D: LASSO Cross-Validation - No Interaction Terms

```
Best RMSE achieved at log(lambda)= -3.1000000000000245

  with average RMSE = 0.7006446013682026 and SE= 0.03449893054191631
```

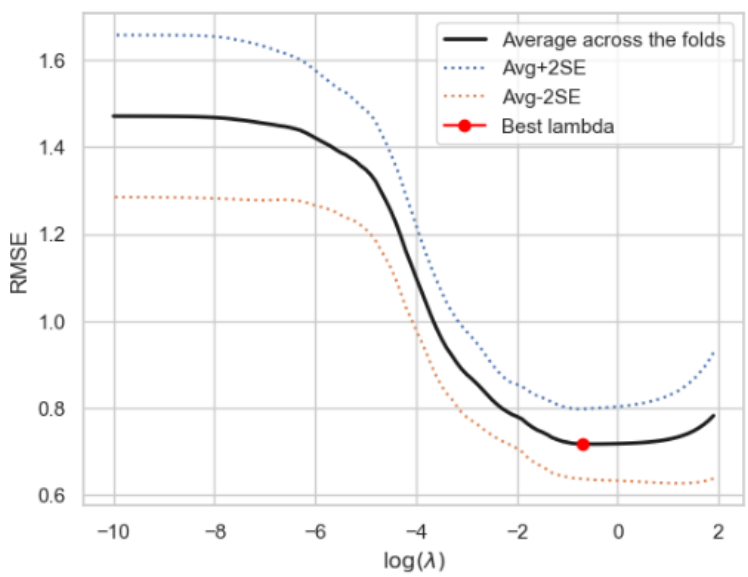


	Features	Coefficient
7	F8	-0.047705
51	F52	-0.011083
54	F55	-0.150009
63	F64	0.015642
149	F150	-0.096732
172	F173	0.394549
173	intercept	0.723277

Appendix E: LASSO Cross-Validation - With Interaction Terms

```
Best RMSE achieved at log(lambda)= -0.700000000000033

  with average RMSE = 0.7167461141412456 and SE= 0.04019124750700586
```



	Features	Coefficient
8202	F55_F173	-0.002749
15223	F173_F173	0.027023
15226	intercept	2.011328

Appendix F: Additional Detail on Linear Regression Model

Least Squares Regression Model: ***outcome***[13] ***~ F8 + F52 + F55 + F64 + F150 + F173***

Features (features in bold below are statistically significant at a 5% significance level):

- **Intercept = 0.7664**
- Negative impact/coefficients:
  a. F8: *Mark the one Springboard value you think... [We, as a company, exhibit most consistently]*" = Mission First; Coefficient = -0.2214
  b. F52: *I am empowered to make decisions within my role* = Strongly agree; Coefficient = -0.1295
  c. **F55: *I am learning & growing my skill set in my current role* = Neutral, Coefficient = -0.4969**
  d. F150: *My preference is to work [x] out of the SF or BLR office* = 1 week/quarter; Coefficient = -0.2687
- Positive impact/coefficients:
  e. F64: *Hiring a diverse group of candidates is a priority to leaders at Springboard* = Neutral; Coefficient = 0.1993
  f. **F173 : *On a scale of 0-10, how likely are you to recommend Springboard to a friend as a place to work?*; Coefficient = 0.3996**
- Performance
  g. Validation RMSE: 0.649077
  h. 5-Fold Cross Validation Average Validation RMSE: 0.7779, Standard Error: 0.0596

[13] "*I see myself working at Springboard a year from now*"; scale of 1 to 5 where Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly agree = 5